Guanhua Zhang

PhD Candidate in Computer Science • University of Stuttgart, Germany
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EDUCATION

University of Stuttgart Sep 2020 – Sep 2024 (Expected)
PhD in Computer Science • Direction: Deep Learning + Human Computer Interaction (HCI)
- Institute for Visualisation and Interactive Systems; Supervised by Prof. Andreas Bulling

International Max Planck Research School for Intelligent Systems (IMPRS-IS) Sep 2020 - Sep 2024 (Expected)
PhD Scholar (Second Affiliation)
- Thesis Advisory Committee: Prof. Andreas Bulling, Prof. Steffen Staab, Prof. Katherine J. Kuchenbecker

Tsinghua University Sep 2017 - Jul 2020
MSc in Computer Science and Technology
- Institute of HCI and Media Integration; Supervised by Prof. Yong-jin Liu

Beijing University of Posts and Telecommunications (BUPT) Sep 2013 - Jun 2017
BSc in Computer Science and Technology

PROFESSIONAL EXPERIENCE

University of Stuttgart, Perceptual User Interfaces Group Stuttgart, Germany
Scientific Researcher (Wissenschaftlicher Mitarbeiterin) Sep 2020 - Sep 2024 (Expected)

User Behaviour Simulation via **Generative AI** (**Diffusion Models**)
- Proposed the first model to **disentangle multiple causal factors that influence user behaviour** and to **generate personalised and realistic action traces,** using a **semi-supervised** diffusion-based autoencoder
- Co-proposed the first model to simulate **personalised eye movements**
- Co-proposed a diffusion model to generate raw **eye movements** on 360º **VR** images
- Simulated **multimodal mobile typing** behaviour (location, size and mobile sensors) via diffusion models

Learning Useful Representations of User Action Sequences to Improve Downstream Tasks
- Designed the first **self-supervised** model learning **lower-dimensional** representations of computer mouse behaviour that can be reused across 4 datasets and 3 downstream tasks (**user identification**, **user task recognition** & **next activity prediction**), leveraging **Transformer-based masked autoencoders** inspired by LLMs
- Used **LLMs** to recognise interaction goals from user actions via prompt engineering and fine-tuning
- The first to use **NLP methods** to investigate a flexible language-like structure in interactive behaviour
- Implemented a prototype for **user intention prediction** via **Bayesian models, Random Forest, SVM, CNN** and **LSTM**
- Conducted **A/B testing** and **statistical analyses** for correlation understanding and performance comparison

In-The-Wild User Data Collection Online System
- Implemented a system to remotely collect mouse and keyboard behaviour while users use their computers, based on **Javascript, Node.js and SQL**

Tsinghua University, Graphics & Geometric Computing Group Beijing, China
Scientific Researcher Dec 2015 – Jul 2020 (Collaboration ongoing)

Emotion Recognition From EEG Using Deep Learning Networks
- Designed a **dynamic graph convolutional network** that automatically selects the most relevant channels by a sparsity-constraint loss
- Co-proposed and evaluated an **attention-based LSTM** with **domain discriminator** to minimise data distribution shift across subjects and sessions
- Proposed multi-target emotion recognition model using feature engineering and Bayesian model**,** SVM**,** Random Forest and LSTM; designed subject-(in)dependent experiments

Chinese Academy of Sciences, Engineering Psychology and Human Factors Laboratory Beijing, China
Research Assistant Sep 2017 – Jun 2020
- Co-designed and conducted psychology studies inducing emotions and collecting multimodal signals (EEG, ECG, GSR, PPG and skin temperature)

Beijing Psychtech Technology Co. Ltd Beijing, China
Applied Scientist Sep 2018 – Jan 2019
- Designed and implemented a **real-time stress and workload detection prototype** based on CNN and MLP, using

multimodal signals (ECG, PPG, skin temperature, salivary cortisol signals and face videos)
- Coordinated with a software engineer team to integrate the prototype into the software product

Beijing University of Posts and Telecommunications, Network Technology Center Beijing, China
Research Assistant Sep 2015 – Feb 2016
- Evaluated an algorithm for mobile service selection and recommendation

LEADERSHIP

Supervisor of 12 Master/Bachelor theses, 9 Internships, 6 Course projects 2017-Now
Teaching assistant & Co-organiser of 5 courses about ML + HCI 2020-Now
Co-Organiser of two Multimediate grand challenges, ACM Multimedia (MM'23 '21) 2021-2023
Vice President & Outstanding Individual of Department Student Union, Tsinghua University 2018-2019

SKILLS

Programming: Python, C/C++, Javascript, HTML, Node.js, SQL, Matlab, Java, Pascal; Bash, Git; LaTeX, Markdown; Qt, ActionScript; Pytorch, Keras, TensorFlow; Pandas, SkLearn; Docker; Cloud Computing
Research: Statistical analysis; A/B testing; Data collection; Experiment design; Writing & presentation
Languages: Chinese (Native), English (C1), German (A2, taking B1 course)

FULL PUBLICATIONS

[1] Guanhua Zhang, Zhiming Hu, Mihai Bâce, Andreas Bulling. *Mouse2Vec: Learning Reusable Semantic Representations of Mouse Behaviour*, In Proc. ACM CHI conference on Human Factors in Computing Systems (**CHI**). 2024.
[2] Guanhua Zhang, Matteo Bortoletto, Zhiming Hu, Lei Shi, Mihai Bâce, and Andreas Bulling. *Exploring Natural Language Processing Methods for Interactive Behaviour Modelling*. In Proc. IFIP TC13 Conference on Human-Computer Interaction (**INTERACT**). 2023. **[Nominated for Best Doctoral Student Paper Award]**
[3] Fang Liu, Pei Yang, Yezhi Shu, Fei Yan, Guanhua Zhang, and Yong-Jin Liu. *Emotion Dictionary Learning with Modality Attentions for Mixed Emotion Exploration*. IEEE Transactions on Affective Computing (**TAFFC, IF=11.2**). 2023.
[4] Guanhua Zhang, Susanne Hindennach, Jan Leusmann, Felix Bühler, Benedict Steuerlein, Sven Mayer, Mihai Bâce, and Andreas Bulling. *Predicting Next Actions and Latent Intents during Text Formatting*. In Proc. the CHI Workshop Computational Approaches for Understanding, Generating, and Adapting User Interfaces. 2022.
[5] Guanhua Zhang, Minjing Yu, Yong-Jin Liu, Guozhen Zhao, and Dan Zhang. SparseDGCNN: Recognizing Emotion from Multichannel EEG Signals. IEEE Transactions on Affective Computing (**TAFFC, IF=11.2**). 2021. **[72 citations]**
[6] Philipp Müller, Dominik Schiller, Dominike Thomas, Guanhua Zhang, Michael Dietz, Patrick Gebhard, Elisabeth André, and Andreas Bulling. *MultiMediate: Multi-modal Group Behaviour Analysis for Artificial Mediation*. In Proc. ACM Multimedia (**MM**). 2021.
[7] Xiaobing Du, Cuixia Ma, Guanhua Zhang, Jinyao Li, Yu-Kun Lai, Guozhen Zhao, Xiaoming Deng, Yong-Jin Liu, and Hongan Wang. *An Efficient LSTM Network for Emotion Recognition from Multichannel EEG Signals*. IEEE Transactions on Affective Computing (**TAFFC, IF=11.2**). 2020. **[155 citations]**
[8] Guozhen Zhao, Yulin Zhang, Guanhua Zhang, Dan Zhang, and Yong-Jin Liu. *Multi-target Positive Emotion Recognition from EEG Signals*. IEEE Transactions on Affective Computing (**TAFFC, IF=11.2**). 2020.
[9] Guanhua Zhang, Minjing Yu, Guo Chen, Yiheng Han, Dan Zhang, Guozhen Zhao, and Yong-Jin Liu. *A Review of EEG Features for Emotion Recognition*. SCIENTIA SINICA Informationis. 2019. **[2021 Hot Paper Award]**
[10] Chengwen Zhang, Lei Zhang, Guanhua Zhang. *QoS-aware mobile service selection algorithm*. Mobile Information Systems. 2016.

Under Review

[11] Chuhan Jiao, Yao Wang, Guanhua Zhang, Mihai Bâce, Zhiming Hu, Andreas Bulling. *DiffGaze: A Diffusion Model for Continuous Gaze Sequence Generation on 360° Images*, IEEE Transactions on Visualization and Computer Graphics (**TVCG**, **IF=5.2**). 2024
+ 2 other anonymous submissions

AWARDS AND HONORS

Gemalto Scholarship, The highest amount comprehensive scholarship in Department 2019
BUPT Merit Student, Top 3% 2014-2016
BUPT Scholarship, Top 5% 2014-2016
2nd Prize at National Olympiad in Informatics in Provinces (Programming & Algorithm), Shandong, China 2012